

Target validation paths — urothelial-mets-her2-discordant-kndl

The diagnostic and biomarker workup that hardens the targetable-feature call

Generated 2026-08-20

Target validation paths

Two orders decide most of this case and neither is a new test. A validated HER2 IHC was drawn in 2026-08 and only its tier is missing from the record; that tier governs whether trastuzumab deruxtecan continues on the tumor-agnostic footing written for IHC 3+, and whether the HER2 ADCs gated at 2+ or 3+ (the disitamab vedotin class) are reachable at all. The second is a records request for sequencing already performed between 2023 and 2025, which carries the FGFR3 and FGFR2 status behind erdafitinib (THOR, NCT03390504), the identity of the single OncoKB-actionable alteration the 2023 report counts without naming, and the ERBB2 copy number. If the tier comes back low and the sequencing returns nothing actionable, both of those branches close, and what remains are research-grade stains that have to be validated before anything can be screened against them.

HER2

Ask for the validated HER2 IHC (clone 4B5 or equivalent, reflex ISH at 2+) reported as a tier with tumor content stated. Retrieval of the 2026-08 report comes before any re-stain, because the stain already exists. Three IHC 3+ results describe 2023 and 2025 tumor, while the only read of the current tumor is a weak research-grade multiplex immunofluorescence signal, which is not a validated HER2 assay and settles nothing in either direction. The 2026-08 nodal core carries 5-10% tumor content and can under-read a membrane stain, so run the 2025-03 recurrence block alongside it with tumor content reported for each. That pairing is what separates real antigen loss from a tumor-poor specimen. ERBB2 copy number sits unnamed among the 12 and 13 copy-number alterations counted in the 2023 reports and refines the tier call; it rides on the same retrieval and on the next plasma draw. Serial tumor-informed ctDNA on both existing platforms (Signatera, NeXT Personal) is the other half of the continue-or-stop question on trastuzumab deruxtecan, and the conflicting 2026-06 and 2026-07 zero draws need reconciling against the source reports before the kinetics mean anything.

ARID1A / TERT promoter / TP53

The retrieval of the full sequencing reports is the highest-yield order in this workup and costs no tissue. Roughly 15-20% of metastatic urothelial carcinoma carries an FGFR3 alteration, and that status gates erdafitinib (NCT03390504); the derived record does not contain it. The same request closes the NTRK, RET and BRAF codon 600 questions behind larotrectinib, entrectinib, repotrectinib, selpercatinib and dabrafenib plus trametinib, returns somatic homologous-recombination status next to the ARID1A synthetic-lethality hypothesis, and supplies the exome data for the TMB recomputation and the HLA analysis below. Expect most of those to read negative. That is the normal result for rare alterations in this disease, and the case for reading them is that the answers already exist. Two practical caveats: an FGFR3 alteration found on a laboratory-developed panel may still need confirmation on the approved companion assay before treatment, and a DNA-only panel reporting no structural variants does not close NTRK, whose breakpoints sit in large introns. If the reports cannot be released, a comprehensive plasma panel answers the FGFR3 question without touching tissue, though fusion coverage has to be checked before a negative is trusted.

Nectin-4

Both stains, 95-100% membranous, predate 25 months of enfortumab vedotin, and antigen downregulation is a recognized route to enfortumab resistance, so the historical result cannot carry a re-targeting decision now. The 2026-06 Nectin-4-directed imaging study has already been performed, and retrieving that read answers most of the question without spending tissue; a contemporary IHC on post-enfortumab tissue is the fallback. Retained

expression keeps next-generation non-MMAE Nectin-4 agents live, and loss closes the axis cleanly rather than leaving it open as a standing maybe. Even with retained expression, MMAE re-exposure runs against the established neuropathy, so the practical question is whether a non-MMAE agent is reachable, and prior enfortumab is an exclusion in several Nectin-4 protocols regardless of expression.

TROP2

The validated TROP2 IHC was also performed in 2026-08 and, again, only the tier is missing, so retrieval comes first. Ask for it as a tier (percent membranous staining times intensity, or an H-score) rather than as positive or negative. The weak multiplex signal is a low positive and not a negative, and low expression predicts TROP2-directed ADC benefit less reliably than strong expression does. That is what the number changes: it sets the confidence on the datopotamab deruxtecan and sacituzumab tirumotecan class rather than opening or closing it, since TROP2-ADC development in urothelial carcinoma has not been strictly expression-gated.

PRAME and HLA-A*02:01

*HLA-A*02:01 is confirmed, so expression is the only gate left on the PRAME ImmTAC and TCR-T class, and those protocols screen on validated IHC (clone EPR20330) or RT-PCR at a stated threshold rather than on the research imaging performed in 2026-08. RT-PCR tolerates the 5-10% tumor content of the nodal cores better than IHC scoring does. This is the gate on IMA203 via NCT03686124, the solid-tumor study, and on brenetafusp via NCT04262466. The other registered IMA203 study is restricted to cutaneous melanoma and does not apply here. NY-ESO-1 (CTAG1B) is the second A*02:01-restricted antigen with an open TCR-T bench and should be run off the same block in the same order. Separately, tumor HLA class I loss of heterozygosity can be computed from the existing tumor and normal exome data; somatic loss of A*02:01 would blunt every HLA-restricted approach here, the neoantigen vaccine programs included. That computation is research-grade and would not satisfy protocol eligibility on its own.*

Mesothelin

Strong mesothelin in urothelial carcinoma is unusual enough that it either opens a serious cell-therapy option or turns out to be an artifact of an unvalidated research assay, and only a validated clinical stain (clone 5B2 or the candidate protocol's own assay, tumor content stated) separates those. Most mesothelin cell-therapy protocols run their own central assay at screening, so a local positive opens the door rather than satisfying it, and pleural and pericardial normal expression is the on-target risk a sponsor will weigh next. A serum soluble mesothelin-related peptide level is a low-priority add-on that rides a routine draw and flags a soluble-antigen sink before a CAR-T or engager protocol. Renal impairment raises that level independently of tumor, and creatinine and eGFR are absent from the record, so read any elevation against a current renal panel.

MUC1

MUC1 sits behind mesothelin: the same research assay, the same missing validation, a thinner therapeutic bench. Normal epithelium carries wild-type MUC1, so a strong signal only means something once the assay resolves the tumor-associated glycoform the candidate agent targets, Tn-MUC1 where that is the epitope, using the clone the protocol names. One slide on the block already being cut answers it. Until then the signal is a lead and not a result.

B7-H3

B7-H3 (CD276) has never been assessed in this patient, and it is the pan-cancer surface target with the deepest current trial bench. One extra slide on the cut already planned for HER2 and TROP2 is the only reason it earns tissue in a case this tissue-poor. Expression would open solid-tumor ADC and bispecific cohorts of the ifinatamab deruxtecan class, though sponsors differ on B7-H3 IHC standardization and a protocol may still require its own assay at screening, and a deruxtecan-payload agent would have to be weighed against the unresolved pulmonary findings.

TMB

The 12.4 mut/Mb figure came off an institutional 468-gene laboratory-developed panel, while the tumor-agnostic indication reads TMB-H by an FDA-approved test. Recomputing from the whole-exome data already generated costs no tissue and settles whether the number survives a cross-assay comparison; report the caller and the filtering rather than the bare number, because panel-derived and exome-derived values differ systematically. PD-L1 by a companion clone (22C3 pharmDx or equivalent) is worth repeating only once a specific protocol names a clone and a cutoff: the two results on file, CPS 5 on the primary against CPS 0 in the liver metastasis, are discordance rather than a number. Immune-infiltrate density (CD3, CD8, macrophages) can be re-queried from the 2026-08 multiplex run already performed, if that panel carried T-cell and myeloid markers, with post-SBRT inflammation stated as a caveat.

Personalized neoantigens

An IFN-gamma EliSpot on post-mRNA-series PBMCs against the 24 predicted neoepitope pools, with CD4 and CD8 deconvolution against a matched pre-vaccine timepoint, is the next readout on this axis. The only immunogenicity data so far is 2 of 14 evaluable pools at 4 months, CD4-skewed, with several pools not evaluable, which is too thin to say whether the vaccine axis is doing anything. Banked serial PBMCs mean no new collection. Pool-level results cannot name the peptide that drove a response, so plan deconvolution for any positive pool if the result is meant to guide an expansion product, and read it against the HLA loss-of-heterozygosity analysis: presentation loss would explain a flat result better than a failed vaccine would.

FAP

The one FAPI PET on record found no FAP-avid urothelial disease and lit up granulomatous prostate tissue instead, a classic false positive, so stromal positivity on an unvalidated multiplex assay does not make this a treatment lead. The 2026-06 repeat study has already been performed and its read costs nothing to retrieve. Recent SBRT to the nodal disease adds treatment-related fibroblast activation, which makes new uptake harder to attribute to tumor stroma. FAPI PET is not FDA approved and is available in the US mainly on research protocols, so a repeat study means an enrolling site rather than a clinical order.

Biomarkers measured, but not to decision resolution

Nine markers on this list have a result on file that cannot carry a treatment decision as it stands. The target call is not what is in doubt for these; the resolution of the existing result is. A reader who conflates the two will think a biomarker is unproven when it is only under-measured. Each one has a matching order in the tables below, so none of them was dropped.

- **HER2.** Three validated IHC 3+ results, all on 2023 and 2025 tissue; the current tumor has only a weak research-grade multiplex read. The validated 2026-08 IHC tier, reported with tumor content, closes it.

- **TROP2.** A weak multiplex signal recorded as low expression, not as a negative. The validated 2026-08 IHC reported as a tier closes it.
- **Nectin-4.** 95-100% membranous on two specimens, both predating 25 months of enfortumab vedotin. The 2026-06 imaging read, or a contemporary IHC on post-enfortumab tissue, closes it.
- **Mesothelin.** Strong on research-grade multiplex immunofluorescence only. A validated clinical mesothelin IHC with tumor content stated closes it.
- **MUC1.** Strong on the same research assay, with no validated tier and no glycoform resolution. A validated MUC1 IHC using the clone and threshold a candidate protocol names closes it.
- **PRAME.** Imaging was performed in 2026-08 and the read is not in the record; imaging is not what a PRAME protocol screens on. Validated PRAME IHC (EPR20330) or RT-PCR at the protocol's threshold closes it.
- **TMB.** 12.4 mut/Mb from a laboratory-developed panel against a label written for an FDA-approved test. TMB recomputed from the existing whole-exome data, or a companion-diagnostic-grade report, closes it.
- **PD-L1.** CPS 5 on the primary against CPS 0 in the liver metastasis, both from 2023, on a laboratory clone validated as comparable to 22C3 rather than the companion assay itself. A companion-assay read on contemporary tissue closes it, and only when a protocol names a clone and a cutoff.
- **FAP.** Stromal positivity on an unvalidated multiplex assay against a FAPI PET that showed no avid urothelial lesion. Quantitative FAPI uptake in the known nodal disease, read with the post-SBRT caveat, closes it.

Where to order these assays

Assay	Provider	Decision gated	Contact
Validated HER2 IHC (clone 4B5 or equivalent) with reflex ISH if 2+, on contemporary tumor tissue, reported as an expression tier (3+ / 2+ / 1+ / 0) with tumor content stated	NeoGenomics Laboratories (preferred) (HER2 IHC clone 4B5 (APH-HERX-01AX) with HER2 FISH reflex)	Continue versus stop trastuzumab deruxtecan, and eligibility for HER2 ADCs gated on IHC 2+/3+ (disitamab vedotin class).	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 866-776-5907 option 3
Validated HER2 IHC (clone 4B5 or equivalent) with reflex ISH if 2+, on contemporary tumor tissue, reported as an expression tier (3+ / 2+ / 1+ / 0) with tumor content stated	Labcorp Oncology (HER2 IHC with reflex FISH (test 483289))	Continue versus stop trastuzumab deruxtecan, and eligibility for HER2 ADCs gated on IHC 2+/3+ (disitamab vedotin class).	test info · 531 South Spring Street, Burlington, NC 27215 · 800-447-5816
Validated HER2 IHC (clone 4B5 or equivalent) with reflex ISH if 2+, on contemporary tumor tissue, reported as an expression tier (3+ / 2+ / 1+ / 0) with tumor content stated	Quest Diagnostics (HER2 IHC (test 30316); IHC with FISH reflex (test 15547))	Continue versus stop trastuzumab deruxtecan, and eligibility for HER2 ADCs gated on IHC 2+/3+ (disitamab vedotin class).	test info · 500 Plaza Drive, Secaucus, NJ 07094 · 866-697-8378
Validated HER2 IHC (clone 4B5 or equivalent) with reflex ISH if 2+, on contemporary tumor tissue, reported as an expression tier (3+ / 2+ / 1+ / 0) with tumor content stated	Yale Pathology Labs	Continue versus stop trastuzumab deruxtecan, and eligibility for HER2 ADCs gated on IHC 2+/3+ (disitamab vedotin class).	test info · 310 Cedar Street LH 108, PO Box 208023, New Haven, CT 06520-8023 · 877-925-3522

Retrieval of the full existing sequencing reports (institutional 468-gene tumor/normal panel 2023-11, two commercial WES/WTS platforms, WGS 2025-03), read for FGFR3 and FGFR2 status and for the identity of the single OncoKB-actionable alteration	Foundation Medicine (preferred) (FoundationOne CDx)	Erdaftinib eligibility via FGFR3/FGFR2 status (THOR, NCT03390504), plus the identity of the one OncoKB-actionable alteration in the tissue reports.	test info · 400 Summer Street, Boston, MA 02210 · 888-988-3639
Retrieval of the full existing sequencing reports (institutional 468-gene tumor/normal panel 2023-11, two commercial WES/WTS platforms, WGS 2025-03), read for FGFR3 and FGFR2 status and for the identity of the single OncoKB-actionable alteration	Caris Life Sciences (MI Cancer Seek (whole exome plus whole transcriptome))	Erdaftinib eligibility via FGFR3/FGFR2 status (THOR, NCT03390504), plus the identity of the one OncoKB-actionable alteration in the tissue reports.	test info · 4610 South 44th Place, Phoenix, AZ 85040 (specimen receiving) · 888-979-8669
Retrieval of the full existing sequencing reports (institutional 468-gene tumor/normal panel 2023-11, two commercial WES/WTS platforms, WGS 2025-03), read for FGFR3 and FGFR2 status and for the identity of the single OncoKB-actionable alteration	Tempus AI (Tempus xT / xR)	Erdaftinib eligibility via FGFR3/FGFR2 status (THOR, NCT03390504), plus the identity of the one OncoKB-actionable alteration in the tissue reports.	test info · 600 West Chicago Avenue, Suite 510, Chicago, IL 60654
Serial tumor-informed ctDNA on the existing two platforms (Signatera, NeXT Personal), with the conflicting 2026-06 and 2026-07 zero-draw results reconciled against the source reports	Natera (preferred) (Signatera)	Continue versus stop trastuzumab deruxtecan, read together with the HER2 tier.	test info · 13011 McCallen Pass, Building A Suite 100, Austin, TX 78753 · 650-489-9050
Serial tumor-informed ctDNA on the existing two platforms (Signatera, NeXT Personal), with the conflicting 2026-06 and 2026-07 zero-draw results reconciled against the source reports	Personalis (NeXT Personal Dx)	Continue versus stop trastuzumab deruxtecan, read together with the HER2 tier.	test info · 6600 Dumbarton Circle, Fremont, CA 94555 · 855-373-7978
Comprehensive plasma ctDNA panel covering FGFR3 and FGFR2 alterations, ERBB2 copy number, and variants acquired under prior therapy	Guardant Health (preferred) (Guardant360 CDx)	Erdaftinib eligibility when the archival reports cannot be retrieved; the same draw returns ERBB2 copy number.	test info · 505 Penobscot Drive, Redwood City, CA 94063 (laboratory) · 855-698-8887
Comprehensive plasma ctDNA panel covering FGFR3 and FGFR2 alterations, ERBB2 copy number, and variants acquired under prior therapy	Foundation Medicine (FoundationOne Liquid CDx)	Erdaftinib eligibility when the archival reports cannot be retrieved; the same draw returns ERBB2 copy number.	test info · 400 Summer Street, Boston, MA 02210 · 888-988-3639

Comprehensive plasma ctDNA panel covering FGFR3 and FGFR2 alterations, ERBB2 copy number, and variants acquired under prior therapy	Tempus AI (Tempus xF)	Erdaftinib eligibility when the archival reports cannot be retrieved; the same draw returns ERBB2 copy number.	test info · 600 West Chicago Avenue, Suite 510, Chicago, IL 60654
PRAME expression by validated IHC (clone EPR20330) or RT-PCR, scored at the threshold the candidate protocol names	NeoGenomics Laboratories (preferred) (PRAME IHC clone EPR20330 (APT-PRAX-01AX))	PRAME-directed ImmTAC / TCR-T entry (IMA203 via NCT06743126 or NCT03686124; brenetafusp via NCT04262466).	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 866-776-5907 option 3
PRAME expression by validated IHC (clone EPR20330) or RT-PCR, scored at the threshold the candidate protocol names	Mayo Clinic Laboratories (PRAME immunostain (test 615794))	PRAME-directed ImmTAC / TCR-T entry (IMA203 via NCT06743126 or NCT03686124; brenetafusp via NCT04262466).	test info · 3050 Superior Drive NW, Rochester, MN 55901 · 800-533-1710
PRAME expression by validated IHC (clone EPR20330) or RT-PCR, scored at the threshold the candidate protocol names	Discovery Life Sciences	PRAME-directed ImmTAC / TCR-T entry (IMA203 via NCT06743126 or NCT03686124; brenetafusp via NCT04262466).	test info · 900 Hudson Way, Huntsville, AL 35806 · 866-838-2798
Contemporary Nectin-4 IHC on post-enfortumab-vedotin tissue, or retrieval of the 2026-06 Nectin-4-directed imaging read	Discovery Life Sciences (preferred)	Whether any Nectin-4 re-targeting stays on the table after 25 months of enfortumab vedotin.	test info · 900 Hudson Way, Huntsville, AL 35806 · 866-838-2798
Contemporary Nectin-4 IHC on post-enfortumab-vedotin tissue, or retrieval of the 2026-06 Nectin-4-directed imaging read	NeoGenomics Laboratories	Whether any Nectin-4 re-targeting stays on the table after 25 months of enfortumab vedotin.	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 866-776-5907 option 3
Contemporary Nectin-4 IHC on post-enfortumab-vedotin tissue, or retrieval of the 2026-06 Nectin-4-directed imaging read	Yale Pathology Labs	Whether any Nectin-4 re-targeting stays on the table after 25 months of enfortumab vedotin.	test info · 310 Cedar Street LH 108, PO Box 208023, New Haven, CT 06520-8023 · 877-925-3522
Validated clinical mesothelin IHC (clone 5B2 or the candidate protocol's own assay), scored with tumor content stated	NeoGenomics Laboratories (preferred) (Mesothelin IHC)	Entry to mesothelin-directed CAR-T, T-cell engager and ADC protocols.	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 866-776-5907 option 3
Validated clinical mesothelin IHC (clone 5B2 or the candidate protocol's own assay), scored with tumor content stated	Labcorp Oncology (Mesothelin IHC (test zzIO-228))	Entry to mesothelin-directed CAR-T, T-cell engager and ADC protocols.	test info · 531 South Spring Street, Burlington, NC 27215 · 800-447-5816
Validated clinical mesothelin IHC (clone 5B2 or the candidate protocol's own assay), scored with tumor content stated	Discovery Life Sciences	Entry to mesothelin-directed CAR-T, T-cell engager and ADC protocols.	test info · 900 Hudson Way, Huntsville, AL 35806 · 866-838-2798
Validated clinical mesothelin IHC (clone 5B2 or the candidate protocol's own assay), scored with tumor content stated	Yale Pathology Labs	Entry to mesothelin-directed CAR-T, T-cell engager and ADC protocols.	test info · 310 Cedar Street LH 108, PO Box 208023, New Haven, CT 06520-8023 · 877-925-3522

TMB recomputed from the whole-exome data already generated, using a validated caller, or read from a companion-diagnostic-grade panel report	Foundation Medicine (preferred) (FoundationOne CDx)	TMB-H eligibility for trials that name an assay, and the strength of the neoantigen argument behind checkpoint re-intensification.	test info · 400 Summer Street, Boston, MA 02210 · 888-988-3639
TMB recomputed from the whole-exome data already generated, using a validated caller, or read from a companion-diagnostic-grade panel report	Caris Life Sciences (MI Cancer Seek)	TMB-H eligibility for trials that name an assay, and the strength of the neoantigen argument behind checkpoint re-intensification.	test info · 4610 South 44th Place, Phoenix, AZ 85040 (specimen receiving) · 888-979-8669
TMB recomputed from the whole-exome data already generated, using a validated caller, or read from a companion-diagnostic-grade panel report	Tempus AI (Tempus xT)	TMB-H eligibility for trials that name an assay, and the strength of the neoantigen argument behind checkpoint re-intensification.	test info · 600 West Chicago Avenue, Suite 510, Chicago, IL 60654
ERBB2 copy number read from the existing tissue NGS reports (12 and 13 copy-number alterations were counted and never itemized), with plasma ERBB2 copy number on the next ctDNA draw	Guardant Health (preferred) (Guardant360 CDx)	Refines the HER2 tier call ahead of the continue-versus-stop decision on T-DXd; not an independent gate.	test info · 505 Penobscot Drive, Redwood City, CA 94063 (laboratory) · 855-698-8887
ERBB2 copy number read from the existing tissue NGS reports (12 and 13 copy-number alterations were counted and never itemized), with plasma ERBB2 copy number on the next ctDNA draw	Foundation Medicine (FoundationOne Liquid CDx)	Refines the HER2 tier call ahead of the continue-versus-stop decision on T-DXd; not an independent gate.	test info · 400 Summer Street, Boston, MA 02210 · 888-988-3639
HER2 IHC on a second site (2025-03 bladder recurrence block) read alongside the 2026-08 nodal core, with tumor content reported for each	NeoGenomics Laboratories (preferred) (HER2 IHC clone 4B5 (APH-HERX-01AX) with HER2 FISH reflex)	Whether a low HER2 read on the node reflects antigen loss or a tumor-poor specimen.	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 866-776-5907 option 3
HER2 IHC on a second site (2025-03 bladder recurrence block) read alongside the 2026-08 nodal core, with tumor content reported for each	Yale Pathology Labs	Whether a low HER2 read on the node reflects antigen loss or a tumor-poor specimen.	test info · 310 Cedar Street LH 108, PO Box 208023, New Haven, CT 06520-8023 · 877-925-3522
Validated TROP2 IHC reported as a tier (percent membranous staining times intensity, or H-score) rather than as positive or negative	Discovery Life Sciences (preferred)	How much weight the TROP2-ADC trial axis carries (datopotamab deruxtecan class, sacituzumab tirumotecan).	test info · 900 Hudson Way, Huntsville, AL 35806 · 866-838-2798

Validated TROP2 IHC reported as a tier (percent membranous staining times intensity, or H-score) rather than as positive or negative	NeoGenomics Laboratories	How much weight the TROP2-ADC trial axis carries (datopotamab deruxtecan class, sacituzumab tirumotecan).	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 866-776-5907 option 3
Validated TROP2 IHC reported as a tier (percent membranous staining times intensity, or H-score) rather than as positive or negative	Yale Pathology Labs	How much weight the TROP2-ADC trial axis carries (datopotamab deruxtecan class, sacituzumab tirumotecan).	test info · 310 Cedar Street LH 108, PO Box 208023, New Haven, CT 06520-8023 · 877-925-3522
Somatic homologous-recombination gene status (BRCA1, BRCA2, ATM, PALB2, RAD51 paralogs) read off the retrieved tissue NGS report; an HRD score only if a protocol asks for one	Myriad Genetics (preferred) (myChoice CDx (BRCA1/2 plus genomic instability score))	DNA-damage-response trial eligibility (ATR and PARP inhibitor protocols) alongside the ARID1A hypothesis.	test info · 322 North 2200 West, Salt Lake City, UT 84116 · 800-469-7423
Validated MUC1 IHC using the clone and threshold the candidate protocol names, glycoform-specific (Tn-MUC1) where the agent targets that epitope	Yale Pathology Labs (preferred)	Screening for MUC1-directed CAR-T, T-cell engager or vaccine protocols.	test info · 310 Cedar Street LH 108, PO Box 208023, New Haven, CT 06520-8023 · 877-925-3522
Validated MUC1 IHC using the clone and threshold the candidate protocol names, glycoform-specific (Tn-MUC1) where the agent targets that epitope	NeoGenomics Laboratories	Screening for MUC1-directed CAR-T, T-cell engager or vaccine protocols.	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 866-776-5907 option 3
Validated MUC1 IHC using the clone and threshold the candidate protocol names, glycoform-specific (Tn-MUC1) where the agent targets that epitope	Discovery Life Sciences	Screening for MUC1-directed CAR-T, T-cell engager or vaccine protocols.	test info · 900 Hudson Way, Huntsville, AL 35806 · 866-838-2798
B7-H3 (CD276) IHC on tumor tissue	Discovery Life Sciences (preferred)	Screening for B7-H3-directed ADC and bispecific solid-tumor cohorts (ifinamab deruxtecan class).	test info · 900 Hudson Way, Huntsville, AL 35806 · 866-838-2798
B7-H3 (CD276) IHC on tumor tissue	NeoGenomics Laboratories	Screening for B7-H3-directed ADC and bispecific solid-tumor cohorts (ifinamab deruxtecan class).	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 866-776-5907 option 3
B7-H3 (CD276) IHC on tumor tissue	Yale Pathology Labs	Screening for B7-H3-directed ADC and bispecific solid-tumor cohorts (ifinamab deruxtecan class).	test info · 310 Cedar Street LH 108, PO Box 208023, New Haven, CT 06520-8023 · 877-925-3522
NY-ESO-1 (CTAG1B) expression by IHC or RT-PCR, run on the same block as PRAME	Discovery Life Sciences (preferred)	NY-ESO-1-directed TCR-T screening (Ietetresgene autoleucel class), ordered alongside PRAME.	test info · 900 Hudson Way, Huntsville, AL 35806 · 866-838-2798

NY-ESO-1 (CTAG1B) expression by IHC or RT-PCR, run on the same block as PRAME	NeoGenomics Laboratories	NY-ESO-1-directed TCR-T screening (letetresgene autoleucel class), ordered alongside PRAME.	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 866-776-5907 option 3
PD-L1 IHC by the companion clone a candidate protocol names (22C3 pharmDx or equivalent), on contemporary tissue	Labcorp Oncology (preferred) (PD-L1 IHC 22C3 pharmDx (test 451852))	Entry to trials that specify a PD-L1 clone and cutoff; not a gate on checkpoint use itself.	test info · 531 South Spring Street, Burlington, NC 27215 · 800-447-5816
PD-L1 IHC by the companion clone a candidate protocol names (22C3 pharmDx or equivalent), on contemporary tissue	NeoGenomics Laboratories (PD-L1 IHC 22C3)	Entry to trials that specify a PD-L1 clone and cutoff; not a gate on checkpoint use itself.	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 866-776-5907 option 3
PD-L1 IHC by the companion clone a candidate protocol names (22C3 pharmDx or equivalent), on contemporary tissue	Quest Diagnostics (PD-L1 22C3 (test 93279 lung, 36260 non-lung))	Entry to trials that specify a PD-L1 clone and cutoff; not a gate on checkpoint use itself.	test info · 500 Plaza Drive, Secaucus, NJ 07094 · 866-697-8378
Serum soluble mesothelin-related peptide (SMRP) immunoassay	ARUP Laboratories (preferred) (Soluble mesothelin-related peptides, MESOMARK (test 0081284))	Refines the mesothelin target call and flags soluble-antigen sink before a mesothelin cell-therapy protocol.	test info · 500 Chipeta Way, Salt Lake City, UT 84108-1221 · 800-522-2787
Serum soluble mesothelin-related peptide (SMRP) immunoassay	Fujirebio Diagnostics	Refines the mesothelin target call and flags soluble-antigen sink before a mesothelin cell-therapy protocol.	test info · 201 Great Valley Parkway, Malvern, PA 19355 · 877-861-7246
Retrieval of the 2026-06 FAP-directed imaging read, with quantitative FAPI PET uptake in the known nodal disease only if that read is uninformative	UCLA Health, Ahmanson Translational Theranostics Division (preferred)	Whether FAP-directed radioligand concepts stay on the list; currently an imaging-research lead rather than a treatment lead.	test info · 855-731-6040
Retrieval of the 2026-06 FAP-directed imaging read, with quantitative FAPI PET uptake in the known nodal disease only if that read is uninformative	Stanford Nuclear Medicine and Molecular Imaging	Whether FAP-directed radioligand concepts stay on the list; currently an imaging-research lead rather than a treatment lead.	test info · 650-723-6855

Biomarker plan

Recommended assay	Rationale	Suggested assay provider	Tissue requirements
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Validated HER2 IHC (clone 4B5 or equivalent) with reflex ISH if 2+, on contemporary tumor tissue, reported as an expression tier (3+ / 2+ / 1+ / 0) with tumor content stated	The validated stain was already drawn in 2026-08 and only its tier is missing from the record, so the first move is retrieving that report rather than cutting a new slide. Three IHC 3+ results describe 2023 and 2025 tumor; the only read of the current tumor is a weak research-grade multiplex immunofluorescence signal, which is not a validated HER2 assay and cannot settle the question either way. Without the tier, a poorly tolerated T-DXd course continues on three-year-old tissue, and a genuine drop to HER2-low moves the drug off its tumor-agnostic label, which is written for IHC 3+, and onto a bystander-payload argument.	NeoGenomics Laboratories (HER2 IHC clone 4B5 (APH-HERX-01AX) with HER2 FISH reflex) · test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 866-776-5907 option 3	Archival FFPE acceptable; already-cut 2026-08 material preferred, otherwise one block cut (2026-08 nodal core or 2025-03 recurrence) with tumor content stated on the report
Retrieval of the full existing sequencing reports (institutional 468-gene tumor/normal panel 2023-11, two commercial WES/WTS platforms, WGS 2025-03), read for FGFR3 and FGFR2 status and for the identity of the single OncoKB-actionable alteration	The record counts exactly one alteration with an OncoKB treatment interpretation and never names it, and FGFR3 status, which gates erdafitinib in roughly 15-20% of metastatic urothelial carcinoma, appears nowhere in the derived profile. Both answers already sit in reports that were generated and never retrieved, so this costs a records request and no tissue. Leaving it unretrieved risks missing the strongest genomic option in the case while scarce tissue is spent on stains for weaker ones.	Foundation Medicine (FoundationOne CDx) · test info · 400 Summer Street, Boston, MA 02210 · 888-988-3639	No specimen; records retrieval only

Serial tumor-informed ctDNA on the existing two platforms (Signatera, NeXT Personal), with the conflicting 2026-06 and 2026-07 zero-draw results reconciled against the source reports	The stop-versus-continue decision on T-DXd rests on ctDNA kinetics alongside the HER2 tier, and the record carries an internal conflict: two 2026-06 and 2026-07 draws read 0 MTM/mL in one table and are listed as pending in another. The two platforms also point in different directions, with NeXT Personal describing a decrease after a transient post-therapy rise while Signatera reads zero. Reconciling the source reports and taking the next scheduled draws separates a real molecular response after SBRT and two T-DXd doses from assay noise, using blood rather than the scarce tissue.	Natera (Signatera) · test info · 13011 McCallen Pass, Building A Suite 100, Austin, TX 78753 · 650-489-9050	10-20 mL whole blood per draw; no tumor tissue
Comprehensive plasma ctDNA panel covering FGFR3 and FGFR2 alterations, ERBB2 copy number, and variants acquired under prior therapy	If the archived reports cannot be released or turn out not to cover FGFR3, a plasma panel answers the erdafitinib question without touching tissue, and detectable ctDNA on two MRD platforms says shed is adequate for it. The same draw returns ERBB2 copy number and anything acquired across 25 months of enfortumab vedotin and pembrolizumab. A negative plasma result is weaker than a negative tissue result, so it argues for tissue testing rather than closing the question.	Guardant Health (Guardant360 CDx) · test info · 505 Penobscot Drive, Redwood City, CA 94063 (laboratory) · 855-698-8887	Two Streck tubes of whole blood (about 20 mL)

PRAME expression by validated IHC (clone EPR20330) or RT-PCR, scored at the threshold the candidate protocol names	HLA-A*02:01 is confirmed, so expression is the only remaining gate on the PRAME ImmTAC and TCR-T class, and those protocols screen on validated IHC or RT-PCR rather than on the research imaging performed in 2026-08. PRAME in urothelial carcinoma is a subset finding rather than the rule, so a negative is a realistic and useful result. The MAGE-A4 sequence is the precedent: HLA matched, expression negative, option closed by a test rather than by assumption.	NeoGenomics Laboratories (PRAME IHC clone EPR20330 (APT-PRAX-01AX)) · test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 866-776-5907 option 3	Archival FFPE block or 5-10 unstained slides; RT-PCR tolerates the 5-10% tumor content of the nodal cores better than IHC scoring
Contemporary Nectin-4 IHC on tissue, or retrieval of the 2026-06 Nectin-4-directed imaging read	Both stains predate 25 months of enfortumab vedotin, and antigen downregulation is a recognized route to EV resistance, so a 95-100% membranous result from 2023 and 2025 cannot carry a re-targeting decision now. The 2026-06 Nectin-4 imaging study has already been performed and its read would answer most of this without touching tissue. Retained expression keeps next-generation non-MMAE Nectin-4 agents on the table; loss closes the axis cleanly instead of leaving it open as a standing maybe.	Discovery Life Sciences · test info · 900 Hudson Way, Huntsville, AL 35806 · 866-838-2798	Archival FFPE acceptable (one slide on the shared block cut); zero tissue if the 2026-06 imaging read is informative

Validated clinical mesothelin IHC (clone 5B2 or the candidate protocol's own assay), scored with tumor content stated	Strong mesothelin in urothelial carcinoma is unusual enough that it either opens a serious cell-therapy option or turns out to be an artifact of an unvalidated research assay, and only a validated stain separates those. Apheresis product and PBMCs are already banked, so a confirmed result would be immediately usable at screening. Screening on the multiplex result alone risks consenting to a protocol the tumor cannot support.	NeoGenomics Laboratories (Mesothelin IHC) · test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 866-776-5907 option 3	Archival FFPE acceptable; one slide from the shared block cut, preferably the 2025-03 recurrence
TMB recomputed from the whole-exome data already generated, using a validated caller, or read from a panel report	The 12.4 mut/Mb value sits above the TMB-H bar but comes from an institutional 468-gene laboratory-developed panel, while the tumor-agnostic indication reads TMB-H by an FDA-approved test, and FoundationOne CDx remains the only approved TMB companion diagnostic. Recomputing from the whole-exome sequencing already performed costs no tissue and settles whether the number survives a cross-assay comparison. It matters less for pembrolizumab, which this patient has already progressed through, than for trials that gate on a named assay and for the neoantigen argument behind the vaccine and ipilimumab-addition strategies.	Foundation Medicine (FoundationOne CDx) · test info · 400 Summer Street, Boston, MA 02210 · 888-988-3639	No specimen if recomputed from existing exome data; archival FFPE if a companion-diagnostic panel is ordered instead

ERBB2 copy number read from the existing tissue NGS reports (12 and 13 copy-number alterations were counted and never itemized), with plasma ERBB2 copy number on the next ctDNA draw	The 2023 panel reported 12 copy-number alterations in the primary and 13 in the liver metastasis without naming them, so whether ERBB2 was amplified is already answered inside a report nobody has retrieved. Amplification underneath the 3+ stain makes true antigen loss less likely and helps interpret a discordant weak multiplex signal; its absence pushes the other way. This rides on the records retrieval and the plasma draw already planned and uses no tissue.	Guardant Health (Guardant360 CDx) · test info · 505 Penobscot Drive, Redwood City, CA 94063 (laboratory) · 855-698-8887	No specimen for the retrieval; the plasma read shares the ctDNA draw already scheduled
HER2 IHC on a second site (2025-03 bladder recurrence block) read alongside the 2026-08 nodal core, with tumor content reported for each	A single nodal core at 5-10% tumor content is a thin basis for calling antigen loss, and this case already shows site-to-site discordance in another marker, with CPS 5 on the primary against CPS 0 in the liver metastasis. Staining two sites separates a real drop in HER2 from a sampling artifact of a tumor-poor node. If both sites read low, the antigen-loss interpretation becomes much harder to argue away, which is what the T-DXd decision turns on.	NeoGenomics Laboratories (HER2 IHC clone 4B5 (APH-HERX-01AX) with HER2 FISH reflex) · test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 866-776-5907 option 3	Archival FFPE acceptable; one additional slide from each of the 2025-03 recurrence block and the 2026-08 nodal core, cut in the same batch

IFN-gamma EliSpot on post-mRNA-series PBMCs against the 24 predicted neoepitope pools, with CD4 and CD8 deconvolution and a matched pre-vaccine timepoint	The only immunogenicity readout so far is 2 of 14 evaluative pools at 4 months, CD4-skewed, with several pools not evaluative, which is too thin to say whether the vaccine axis is doing anything. Serial pre- and post-vaccine PBMCs are already banked, so the assay needs no new collection and no tumor tissue. Deconvoluting CD4 from CD8 responses also tells the TIL and adoptive-cell-therapy groups which epitopes are worth expanding against.	No external provider listed; reads off records or data already generated.	Banked PBMCs, or one 30-50 mL blood draw if fresh cells are preferred; no tumor tissue
Validated TROP2 IHC reported as a tier (percent membranous staining times intensity, or H-score) rather than as positive or negative	The validated stain was performed in 2026-08 and only the tier is missing, so retrieval comes before any re-stain. A weak multiplex signal is a low positive and not a negative, and low expression predicts TROP2-ADC benefit less reliably than strong expression, which is exactly what the number changes about how much weight this axis carries. TROP2-ADC development in urothelial cancer has not been strictly expression-gated, so a low tier sets the confidence rather than closing the option.	Discovery Life Sciences · test info · 900 Hudson Way, Huntsville, AL 35806 · 866-838-2798	Archival FFPE acceptable; no new tissue if the 2026-08 report is retrieved

NTRK1/2/3 fusion status read off the retrieved and NGS reports; an RNA fusion panel only if none of them covered NTRK	This is a records problem and not a testing problem: whole-transcriptome sequencing that would show an NTRK fusion has already been run on two platforms and the reports are not in hand. NTRK fusions are rare in urothelial carcinoma so a negative is the expected answer, and a DNA-only panel reporting no structural variants does not close the question, because NTRK breakpoints sit in large introns. It rides free on the retrieval this case needs anyway.	No external provider listed; reads off records or data already generated.	No specimen for the readout; archival FFPE if an RNA fusion panel is ordered
RET fusion status read off the retrieved or NGS report; an RNA fusion panel only if RET was not covered	RET fusions are rare in urothelial carcinoma and nothing in this record suggests one, so the expected result is negative. It costs nothing beyond the retrieval already needed, and leaving it unread means an approved oral agent stays formally unassessed rather than excluded. A DNA panel without RET intronic coverage would not settle it on its own.	No external provider listed; reads off records or data already generated.	No specimen for the readout; archival FFPE if an RNA fusion panel is ordered
BRAF codon 600 status read off the retrieved full tissue NGS report	BRAF V600E is uncommon in urothelial carcinoma and the honest expectation is that it is absent. It stays on the list because the record counts one OncoKB-actionable alteration without naming it, so the identity of that variant is unknown and BRAF is among the few candidates that would arrive with an approved tumor-agnostic option. Reading it off the existing report costs a records request.	No external provider listed; reads off records or data already generated.	No specimen; readout from the existing report

Somatic gene status (BRCA1, BRCA2, ATM, PALB2, RAD51 paralogs) read off the retrieved tissue NGS report; an HRD score only if a protocol asks for one	Germline testing was negative in 2023-12, which leaves somatic status as the unanswered half, and it sits in a panel report that covers those genes and has never been retrieved. Urothelial carcinoma is not a PARP-inhibitor indication, so the realistic payoff is trial eligibility next to the ARID1A synthetic-lethality hypothesis rather than an approved drug. No new specimen is involved.	Myriad Genetics (myChoice CDx (BRCA1/2 plus genomic instability score)) · test info · 322 North 2200 West, Salt Lake City, UT 84116 · 800-469-7423	No specimen for the readout; archival FFPE if an HRD score is later required
Validated MUC1 IHC using the clone and threshold the candidate protocol names, glycoform-specific (Tn-MUC1) where the agent targets that epitope	MUC1 sits behind mesothelin here: the same research assay, the same missing validation, a thinner therapeutic bench. Normal epithelium carries wild-type MUC1, so a strong signal only means something when the assay resolves the tumor-associated glycoform the candidate agent targets. One slide on the block already being cut answers it; until then the signal is a lead and not a result.	Yale Pathology Labs · test info · 310 Cedar Street LH 108, PO Box 208023, New Haven, CT 06520-8023 · 877-925-3522	Archival FFPE acceptable; one slide on the shared block cut
B7-H3 (CD276) IHC on tumor tissue	B7-H3 has never been looked at in this patient and it is the pan-cancer surface target with the deepest current trial bench. One extra slide from the block already being cut for HER2 and TROP2 answers it, which is the only reason it earns tissue in a case this tissue-poor. Expression would open solid-tumor ADC and bispecific cohorts, though a deruxtecan-payload agent would still have to be weighed against the unresolved pulmonary findings.	Discovery Life Sciences · test info · 900 Hudson Way, Huntsville, AL 35806 · 866-838-2798	Archival FFPE acceptable; one slide on the shared block cut

NY-ESO-1 (CTAG1B) expression by IHC or RT-PCR, run on the same block as PRAME	The HLA gate is already satisfied, and NY-ESO-1 is the second A*02:01-restricted antigen with an open TCR-T bench, so it costs one assay run next to the PRAME order. Expression in urothelial carcinoma is a minority finding, so a negative is the likely and still useful outcome. MAGE-A4 already showed how quickly an expression assay closes one of these axes.	Discovery Life Sciences · test info · 900 Hudson Way, Huntsville, AL 35806 · 866-838-2798	Archival FFPE acceptable; RT-PCR is the more forgiving option at 5-10% tumor content
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PD-L1 IHC by the companion clone a candidate protocol names (22C3 pharmDx or equivalent), on contemporary tissue	The two results on file are CPS 5 on the primary and CPS 0 in the liver metastasis, which is discordance rather than a number, and both date from 2023 on a laboratory clone validated as comparable to 22C3 rather than the companion assay itself. Checkpoint use in metastatic urothelial carcinoma is not PD-L1 gated, and this patient has already had both a deep response and progression on PD-1 blockade, so the practical weight is low. Repeat it when a specific protocol names a clone and a cutoff, not before.	Labcorp Oncology (PD-L1 IHC 22C3 pharmDx (test 451852)) · test info · 531 South Spring Street, Burlington, NC 27215 · 800-447-5816	Archival FFPE acceptable; one slide on the shared block cut when a protocol requires it
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Tumor HLA class I loss of heterozygosity computed from the existing tumor/normal exome data (LOHHLA or equivalent)	Germline typing is on file and positive for A02:01, but somatic loss of that allele in the tumor would blunt every HLA-restricted approach in play here, from PRAME ImmTAC and TCR-T to the personalized neoantigen vaccines. The tumor and normal exome data needed for the calculation already exists, so this is an analysis rather than a specimen request. Skipping it leaves a class-wide escape mechanism unexamined while several A02:01-dependent programs are being pursued.	No external provider listed; reads off records or data already generated.	No specimen; computed from existing tumor and normal exome data
CD3, CD8 and macrophage density re-queried from the 2026-08 multiplex run already performed on the nodal tissue	The multiplex panel has already been run on the 2026-08 node, so immune-infiltrate density can be read from the existing image data at no tissue cost. An inflamed node supports adding CTLA-4 blockade or an IL-15 superagonist after progression on PD-1 maintenance, while an immune-desert pattern argues the problem is priming rather than checkpoint restraint. This is a research-grade readout that informs how the immune-intensification axis is ranked, not protocol eligibility.	No external provider listed; reads off records or data already generated.	No specimen; re-analysis of the 2026-08 multiplex images already acquired

Serum soluble mesothelin-related peptide (SMRP) immunoassay	Shed mesothelin acts as a sink that blunts mesothelin-directed agents, so a soluble level is worth having before a CAR-T or engager protocol, and it needs blood rather than tissue. The assay is cleared for mesothelioma monitoring rather than for urothelial carcinoma, so a level here corroborates or undercuts the tissue call without replacing it. Low value on its own; it earns a line because it rides on a routine draw.	ARUP Laboratories (Soluble mesothelin-related peptides, MESOMARK (test 0081284)) · test info · 500 Chipeta Way, Salt Lake City, UT 84108-1221 · 800-522-2787	3-5 mL serum on a routine draw; no tumor tissue
Retrieval of the 2026-06 FAP-directed imaging read, with quantitative FAPI PET uptake in the known nodal disease only if that read is uninformative	The one FAPI PET on record found no FAP-avid urothelial disease and lit up granulomatous prostate tissue instead, a classic false positive, so stromal positivity on an unvalidated multiplex assay does not make this a treatment lead. The 2026-06 repeat study has already been performed and its read costs nothing to retrieve. Recent SBRT to the nodal disease adds treatment-related fibroblast activation, which makes any new uptake harder to attribute to tumor stroma.	UCLA Health, Ahmanson Translational Theranostics Division · test info · 855-731-6040	No specimen; imaging only

Decision support, not medical advice. Confirm assay availability and current standards with the treating team and the local pathology service.